D39 rank closure is reproduced by a Hecke-free doubling control

2026-10-30. **Verified finite diagnostic; no IC speedup established.**

The frozen six-arm audit completed after checkpoint upload access was supplied. The saved Hecke+Atkin–Lehner rank rises from 52 to 63 on the same 64 columns modulo 11 when either projected equalities or doubling relations are added. All 64 normalized base logs were recovered and checked in each of four arms. The follow-up control reaches the same rank 63 using doubling rows alone, with **zero Hecke rows**. A separate short-Weierstrass implementation verifies all 256 scalar labels from those four solutions.

Exact scope and receipts

Curve: X_0(11)=11a1, y^2+y=x^3-x^2-10x-20 over F367; full group order 385=5*7*11, subgroup order 11, projection P -> [35]P. The original 64-point CM class has D_K=-39 and conductor 6. Its projected points cover all 11 subgroup elements; this is a dense toy configuration.

The unchanged protocol was published in PR #1306 before execution. The first receipt, launch record, stdout/stderr and successful checkpoint receipt retain the actual attempt. Exit status was 0 under the 60-second cap. Source input SHA-256 is `4bcc8eb1905a9ed79a47cc31820e722dc7a6ff87f18bb37aeebafa35e6cc6df9`; the executed checker hash is in the receipt. The launch records the Python runtime, platform and code revision. Elapsed time is diagnostic only.

Arm / Total rows / Rank mod 11 / Nullity / Replayed normalized base logs

Hecke / 48 / 40 / 24 / not attempted

Hecke + Atkin–Lehner baseline / 112 / 52 / 12 / not attempted

Baseline + projected equalities/zeros / 166 / 63 / 1 / 64/64

Baseline + doubling / 508 / 63 / 1 / 64/64

Baseline + doubling/tripling / 880 / 63 / 1 / 64/64

Baseline + all additions / 934 / 63 / 1 / 64/64

Dense elimination agrees with the frozen independent sparse checker in every arm. All raw points and full-group Hecke identities pass; all projected Hecke/Atkin–Lehner and added relations pass group checks. The small-field point count independently confirms order 385. The audit reports 448 group-add calls for combined doubling/tripling collection and 10,134 including setup and verification; these are implementation counters, not a fair end-to-end comparison or a calibrated cost model.

Hecke-free control and independent replay

The control protocol was fixed locally after the first result and is first published with its outcome. It is an exploratory control, not an independently verifiable holdout. Its receipt and launch retain exit 0 and hashes.

Control, with no Hecke or Atkin–Lehner rows / Rows / Rank mod 11 / Nullity

Projected equalities/zeros only / 54 / 54 / 10

Doubling only / 396 / 63 / 1

Doubling + tripling only / 768 / 63 / 1

The independent replay uses X=x-1/3, Y=y+1/2, giving Y^2=X^3+112X+55 over F367, a separate short-model addition formula and repeated addition instead of the producer's generalized-model double-and-add. It checks each projected coordinate, all new control rows and the 256 prior scalar labels. Anchor column 0 defines the reference generator; no unknown external-target log or target-descent algorithm is supplied.

Interpretation: the new rows close the finite relation gap, but the same closure is available without Hecke information. Accordingly this is not an improvement over the Hecke-free closure baseline and not a pipeline speedup. The tiny subgroup permits many projected collisions; the result does not establish comparable relation yield at cryptographic sizes.

Status and next discriminator

Requirement / Status

Required checkpoint before and after execution / verified complete

Frozen six-arm diagnostic / verified complete

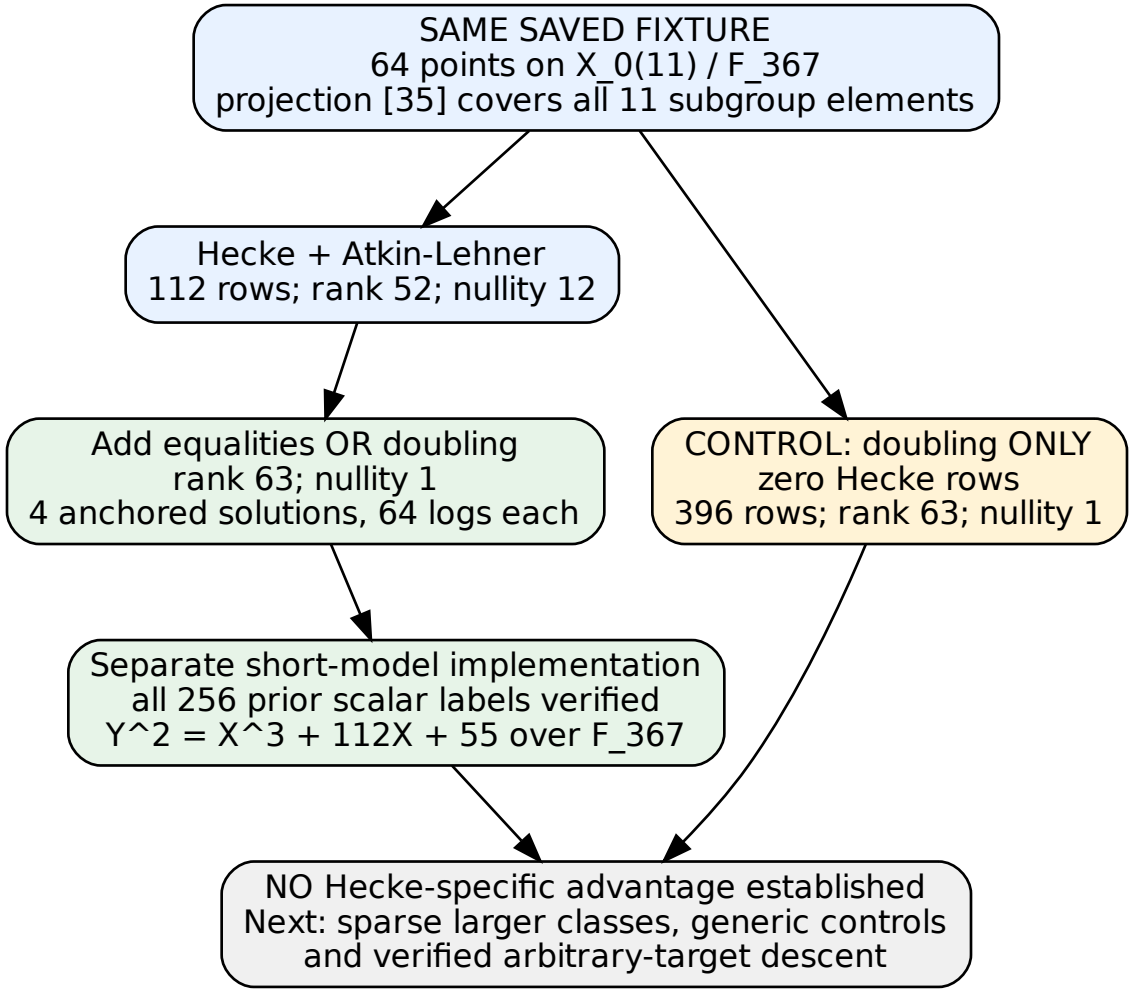
Independent rank checks and anchored replay / verified complete

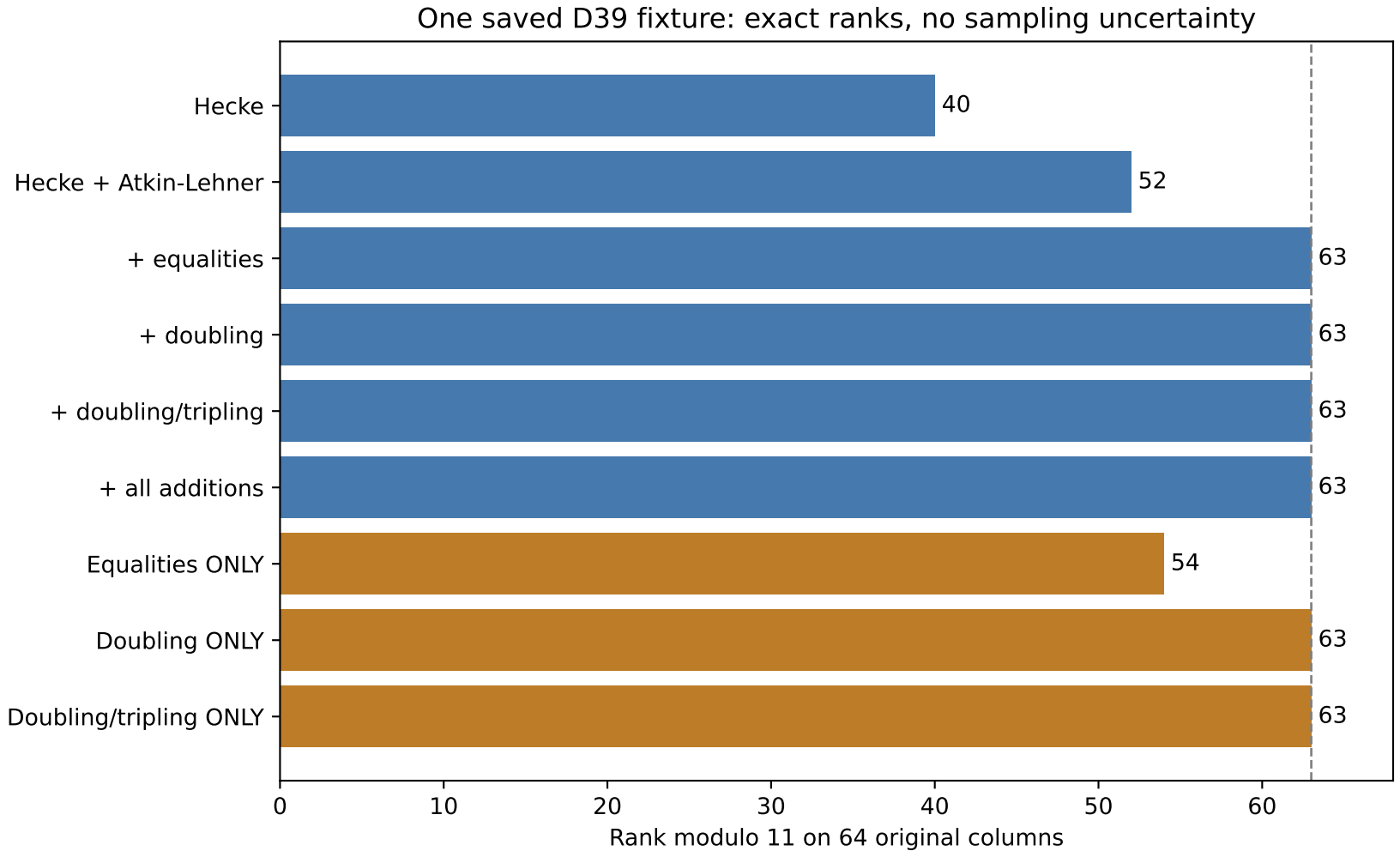
Hecke-free control and independent coordinate/group replay / verified complete

Arbitrary-target descent, scalable advantage and matched rho comparison / unresolved

The previous blocked report and its PDF are preserved as historical evidence. The access problem is resolved for this session. The remaining sparse-closure and descent problem states the next acceptance criteria: larger pinned classes with sparse, predominantly distinct projections, matched generic scalar-closure controls, and verified target descent. No larger-class run was launched in this change.

The new diagram, rank chart and PDF are generated from editable sources. Rebuild with `python3 build_result.py`. The original D39 schematic and canonical surface-transfer graphs were reviewed again. This adds a relation-stage result, not a new isogeny edge; those existing graphs remain unchanged. Original inputs and all earlier evidence are preserved.


SAME SAVED FIXTURE
64 points on X_0(11) / F_367
projection [35] covers all 11 subgroup elements
Hecke + Atkin-Lehner
112 rows; rank 52; nullity 12
Add equalities OR doubling
rank 63; nullity 1
4 anchored solutions, 64 logs each
CONTROL: doubling ONLY
zero Hecke rows
396 rows; rank 63; nullity 1
Separate short-model implementation
all 256 prior scalar labels verified
Y^2 = X^3 + 112X + 55 over F_367
NO Hecke-specific advantage established
Next: sparse larger classes, generic controls
and verified arbitrary-target descent


One saved D39 fixture: exact ranks, no sampling uncertainty
Hecke
Hecke + Atkin-Lehner
+ equalities
+ doubling
+ doubling/tripling
+ all additions
Equalities ONLY
Doubling ONLY
Doubling/tripling ONLY
40
52
63
63
63
63
54
63
63
0
10
20
30
40
50
60
Rank modulo 11 on 64 original columns